

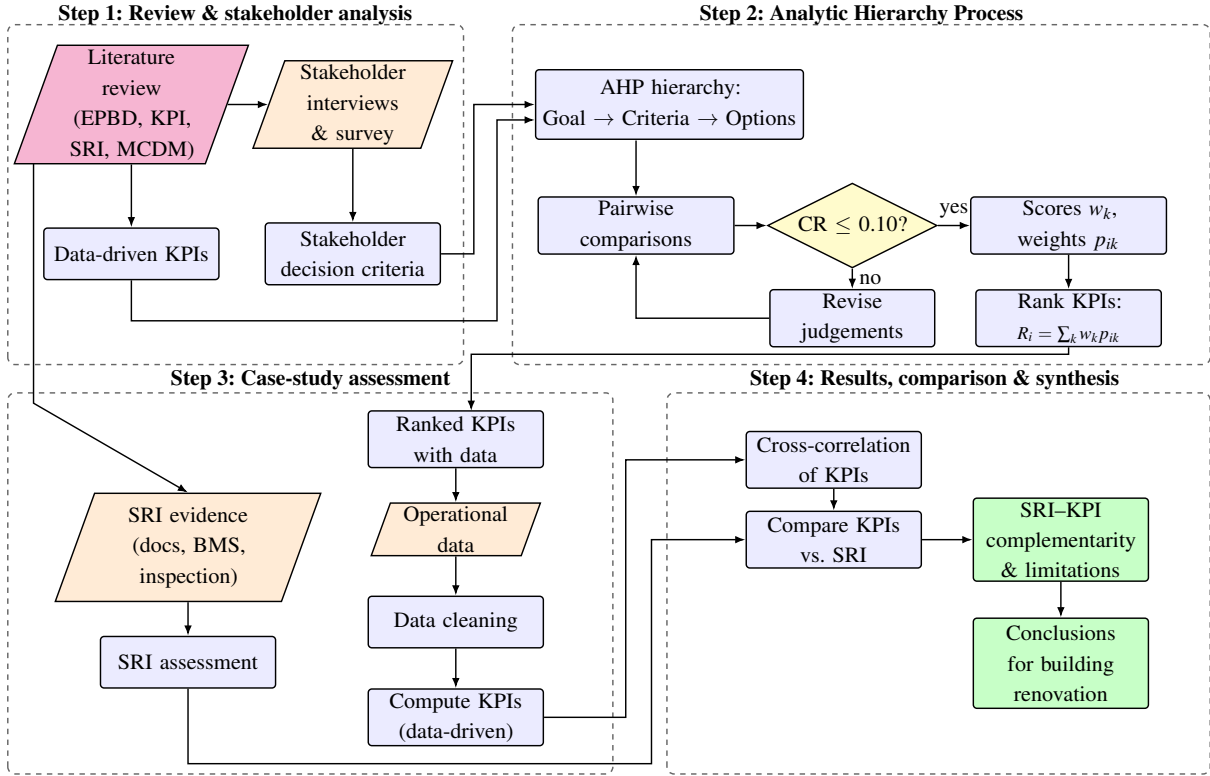


Figure 1: Workflow of the stakeholder-oriented KPI selection method.

2.1 Analytic Hierarchy Process

To support the complex decision-making process of identifying the most relevant KPIs for non-residential building renovation, this studies employs the AHP, a well structured and established technique amongst MCDM, able to perform a consistent multi-criteria and multi-domain assessment. AHP is a three-step method, that starts from 1) the identification of a set of options (KPIs) and decision criteria (Stakeholder's criteria), 2) the assignation of priority score and weights through pairwise comparison, and 3) calculation of the overall scores to find the most suitable option.

1. Hierarchy tree AHP decomposes the problem into its fundamental constituents, that are then organised into a hierarchical structure, with the Goal at the highest level, followed by Criteria and if applicable, the sub-criteria, while the lowest rank contains the Options. In Figure 2 and Table 5 we applied the hierarchy system to our case, defining criteria based on the results of the stakeholders surveys described in subsection 2.3, while the KPIs represents the available options, drawn from the outcomes of subsubsection 2.2.2 and subsubsection 2.2.1.

2. Pairwise comparisons Through a pairwise comparison of the criteria the decision maker rates the relative priority of the criteria grading on a scale from 1 (equal importance) to 9 (extreme importance) the more important criterion, whereas the reciprocal of this value is assigned to the other criterion in the pair. The pairwise comparisons form a positive reciprocal matrix $\mathbf{A} = [a_{ij}]$, where a_{ij} expresses the relative importance of element i over element j . The reciprocal property is defined as:

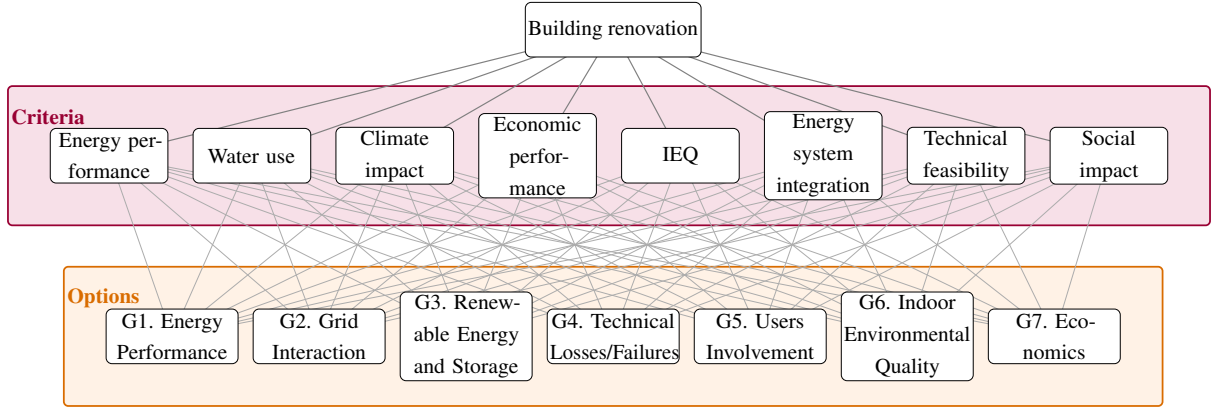


Figure 2: AHP structure for indicator selection.

$$a_{ji} = \frac{1}{a_{ij}}, \quad a_{ii} = 1, \quad i, j = 1, \dots, n, \quad (1)$$

where n is the number of elements compared at a given level of the hierarchy. By normalising each grade by the sum of its respective column, each entry a_{ij} becomes a weighting factor $\tilde{a}_{ij}$.

$$\tilde{a}_{ij} = \frac{a_{ij}}{\sum_{k=1}^n a_{kj}}, \quad (2)$$

The weight vector for each criteria k was then calculated as the average of the normalized values:

$$w_k = \frac{1}{n} \sum_{j=1}^n \tilde{a}_{kj}, \quad \sum_{k=1}^n w_k = 1. \quad (3)$$

To evaluate how well one option i meet each criterion k , respectively to the alternative in the pair, the same process is repeated, obtaining an option score vector p_{ik} that satisfy:

$$\sum_{i=1}^m p_{ik} = 1, \quad (4)$$

where m is the number of KPIs evaluated under that criterion.

The consistency of the decision maker's judgments on criteria and options was assessed employing Saaty's Consistency Ratio (CR) [56, 44]. The scores are considered acceptable if the consistency ratio don't exceed 10%, otherwise it is recommended to revise and adjust judgements to improve the quality. Mathematically, CR is based on the solution of an eigenvalue problem. For each $n \times n$ reciprocal pairwise-comparison matrix $\mathbf{A}$, the normalized weighting vector $\mathbf{w}$ was obtained from the principal right eigenvector. The relationship between the pairwise-comparison matrix $\mathbf{A}$, the weighting vector w_i and p_{ik} , and the principal eigenvalue $\lambda_{\max}$ was expressed as

$$\mathbf{A}\mathbf{w} = \mathbf{s} = \lambda_{\max}\mathbf{w}, \quad (5)$$

Where the $\lambda_{\max}$ was estimated as the mean of the component-wise ratios between the weighted-sum vector and the weighting vector:

$$\lambda_{\max} = \frac{1}{n} \sum_{i=1}^n \frac{(\mathbf{A}\mathbf{w})_i}{w_i} = \sum_{i=1}^n \frac{\sum_{j=1}^n a_{ij}w_j}{nw_i}. \quad (6)$$

The Consistency Index (CI) was calculated as:

$$CI = \frac{\lambda_{\max} - n}{n - 1}. \quad (7)$$

The Consistency Ratio was then determined by:

$$CR = \frac{CI}{RI_n}, \quad (8)$$

where RI_n is Saaty's Random Index for a matrix of order n . Matrices for which $CR > 0.10$ were reviewed and the associated pairwise judgements were revised before deriving the final criterion weights and KPI priorities.

3. Rank the options Finally, a global option ranking R_i is obtained by the mean of a simple weighted summation between the criterion weights w_k and the option scores p_{ik} . A high R_i value indicates that the KPI i is a suitable solution for the decision-making process.

$$P_i = \sum_{k=1}^q w_k p_{ik}, \quad (9)$$

2.2 Performance indicators

2.2.1 Data-driven indicators

A total of 54 data-driven indicators, tailored to describe building performance across seven domains: energy performance (G1), grid integration and interactions (G2), RES and storage (G3), technical losses and failures (G4), user involvement (G5), indoor environmental quality (G6) and economics (G7). The full list of KPIs, correlated with necessary information for their application (e.g. mathematical definition and data requirements) is reported in Table 6. It lies outside of the scope of this work to develop new indicators or to characterise the missing mathematical equation. Through indicators like Primary Energy or Equivalent CO_2 Emissions, G1 achieves the overall objective of describing the energy performance on a broad level, ensuring easy comparison with other buildings. G2 measures a building's ability to utilise smart energy management capabilities offered by the available technologies for flexible operations (i.e. load shifting and shaving) The metrics in G3 assess the performance related to the implementation of RES and storage systems to reduce dependency from the energy grid, towards more energy independent buildings. Building electrification and the use of RES requires a study of the energy system reliability, that can be measured through indicators in G4. G5 assess building performance in relation to system interruptions that can affect user activities and wellbeing, and ultimately improve maintenance costs. Indoor environments are measured with indicators grouped in G6, ensuring acceptable comfort level for the users. Finally G7 implement metrics to measure the introduction of new technologies and control strategies on an economic plane, evaluating the convenience of the proposed solutions.

2.2.2 Smart readiness indicator

The SRI provides a standardised, EU-wide framework in which buildings of very different type and age can be compared on a common basis. To align the performance assessment with the obligations of the EPBD, this study will adopt the official EU Smart Readiness Indicator methodology [18, 57], and the smart readiness of the campus buildings was assessed through the SRI calculation spreadsheet tool (version 4.5).